



Open-Motion

User Manual

Product	Open-Motion blood-flow monitor
Application version	1.4.0
Document revision	July 2026 — draft
Audience	Operators

DRAFT — not a controlled document. This manual is a documentation preview generated from Open-Motion 1.4.0 running on live hardware. It is not a validated Instructions-For-Use document — always defer to official Openwater labeling and training.

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User Manual

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1. About the system

Open-Motion is Openwater's optical blood-flow monitor. Two forehead **sensor modules** (up to 8 cameras each) and a **console** containing a near-infrared laser measure laser-speckle images through the skin and compute, in real time:

- **BFI — Blood Flow Index:** relative index of blood flow (white trace).
- **BVI — Blood Volume Index:** relative index of blood volume (blue trace).

The **Open-Motion** application presents a simple, guided workflow: connect, check sensor contact, scan, review. The display shows large LEFT / RIGHT BFI and BVI readouts backed by one live plot per side, and every scan is checked for sensor contact quality before it starts. Laser safety is enforced by a hardware interlock inside the console, independently of this software.

2. Starting the application

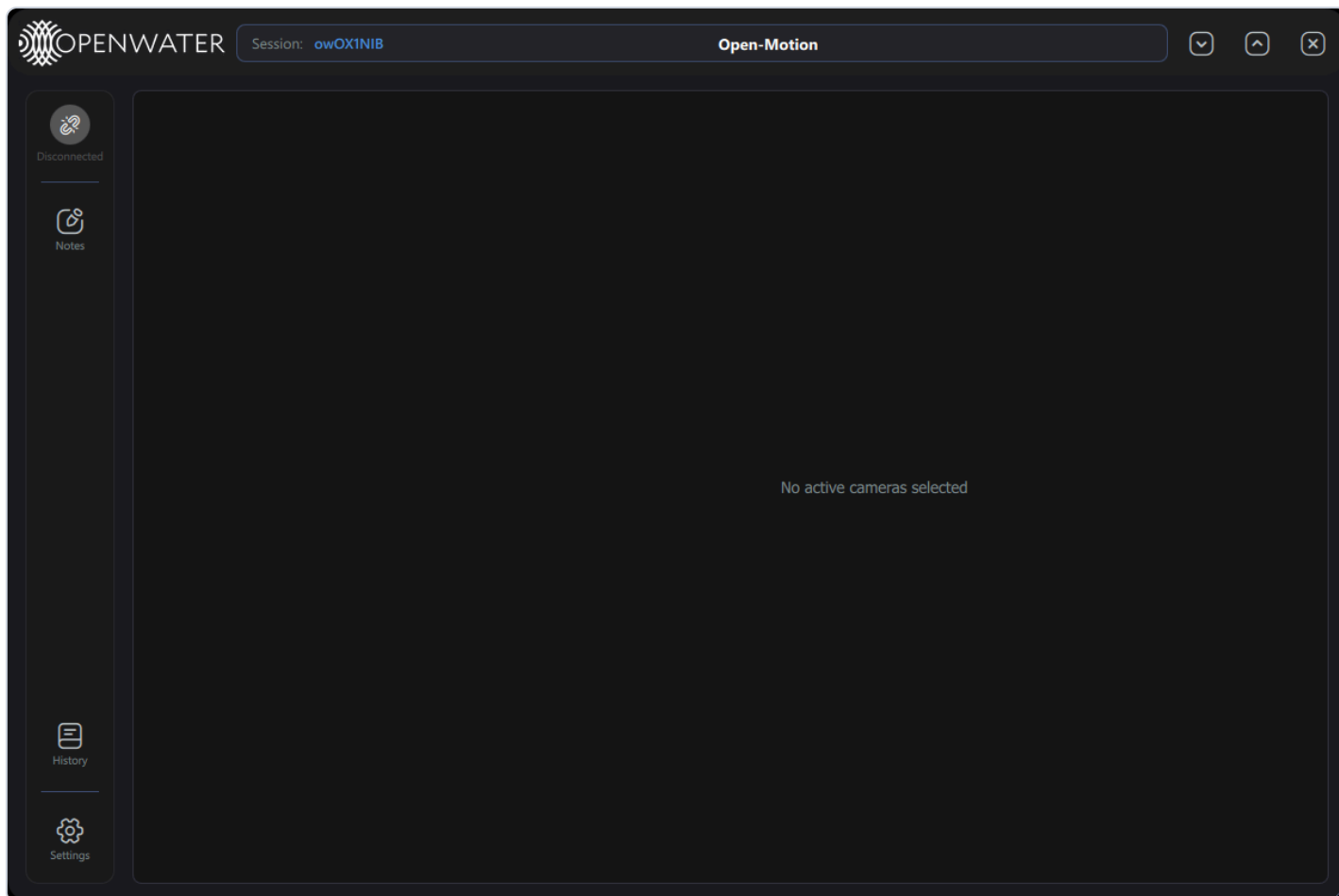
Launch **Open-Motion** from the Start menu or desktop shortcut.

- Only one copy can run at a time. If a second copy is launched, a dialog appears: *"Another instance of the application is already running. Please close the existing instance..."* — close the other copy first.
- The app connects to the console and sensors **automatically** whenever they are plugged in and powered; there is no Connect button.
- After the devices connect, the system performs a brief sensor initialization. Allow the system to settle for about two minutes after connecting before starting a scan.

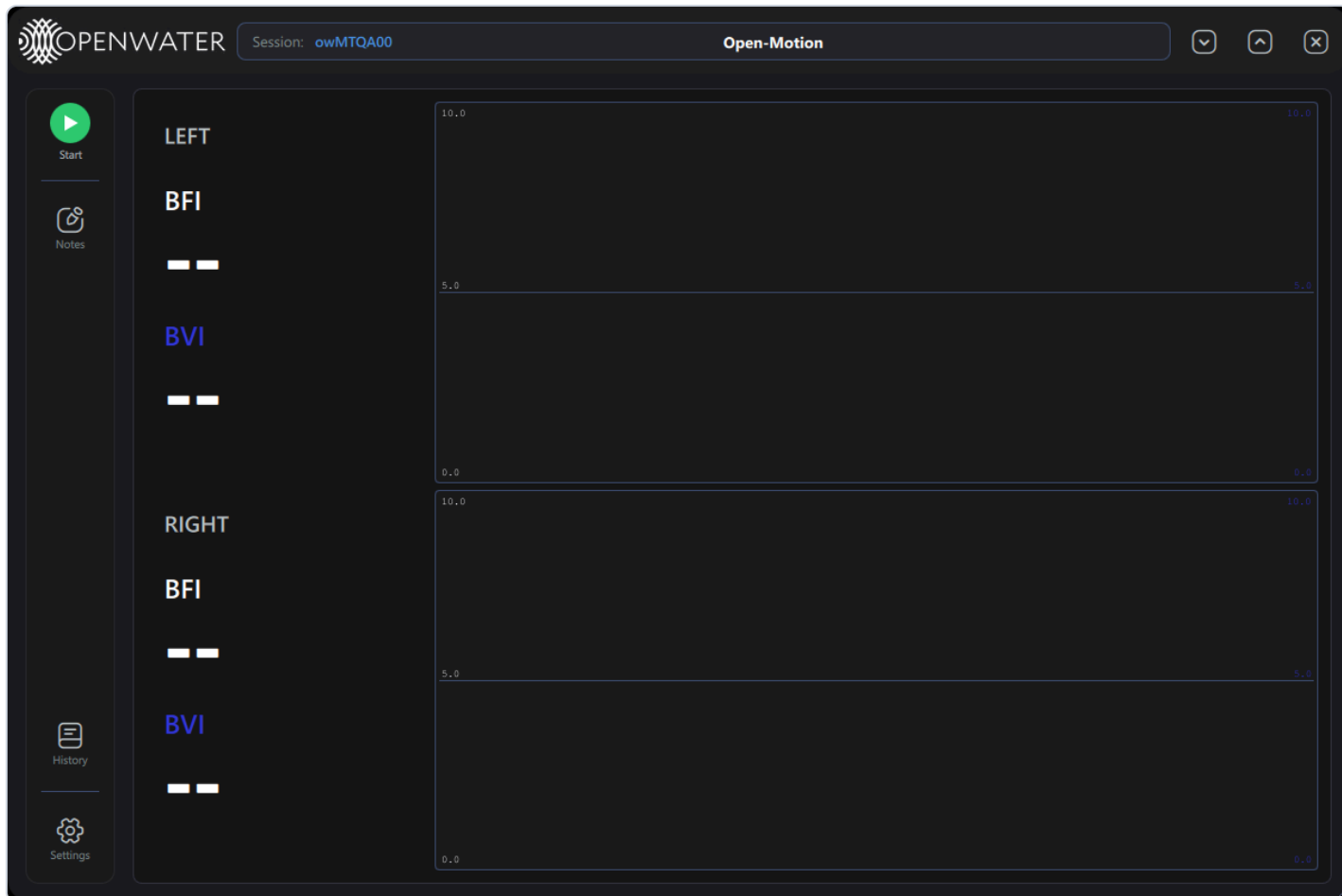
2.1 Connection states

The round badge at the top of the left toolbar tells you the system state at a glance:

Badge	Label	Meaning
Grey circle, chain-link icon	Disconnected	Console and/or sensors not detected. Scanning is disabled.
Green circle, play icon	Start	Everything connected and initialized — ready to scan.
Yellow circle	Start	Start pressed; the system is finishing the previous operation before the scan begins (normally a few seconds).
Red circle, stop icon	Stop	A scan is running. Press to stop.

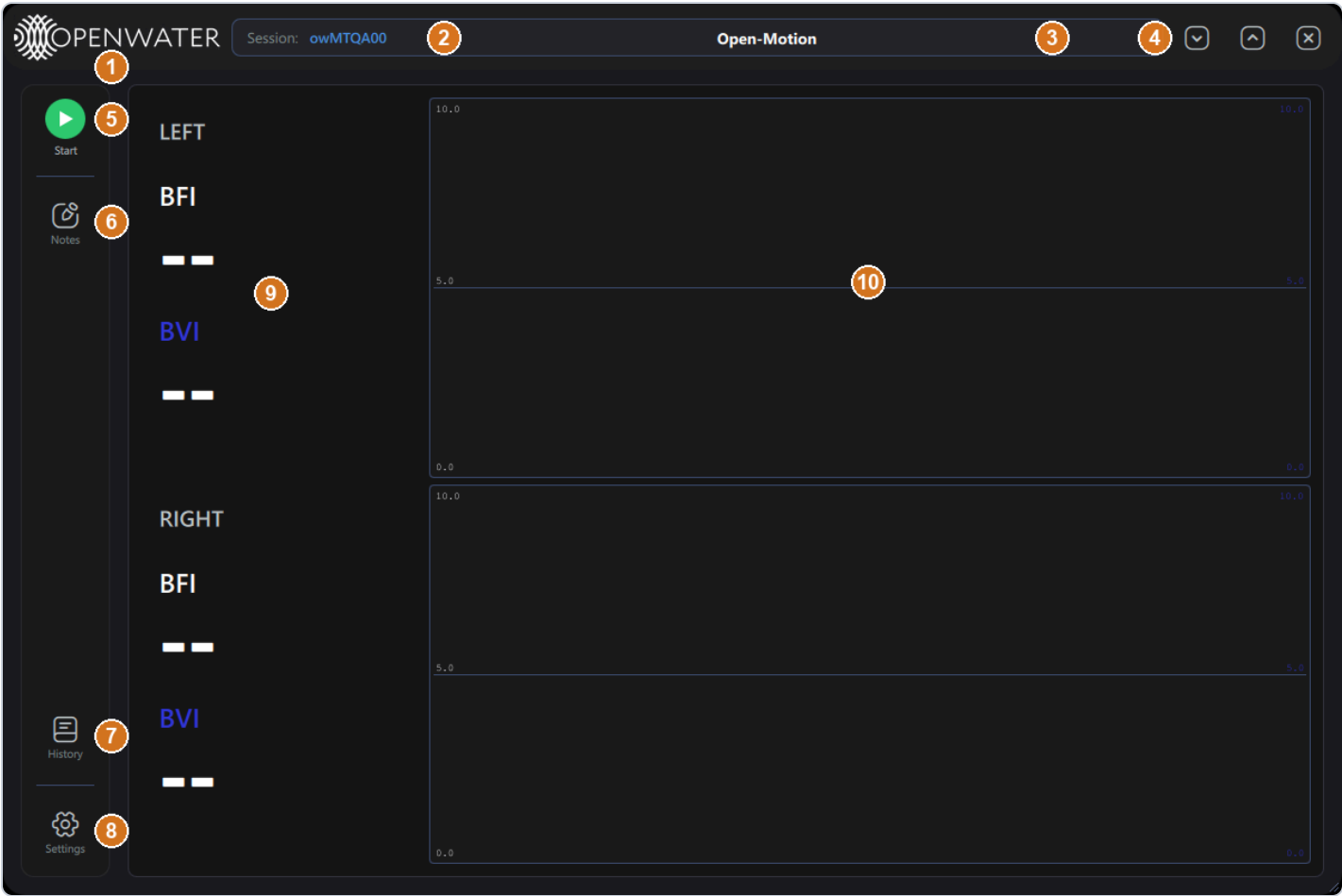


Disconnected: the plot area reads "No active cameras selected" and the badge is grey.



Ready: green Start badge, LEFT/RIGHT panels showing -- (no data yet).

3. The main screen



#	Element	What it does
1	Openwater logo	Branding.
2	Session:	The current session identifier. A new identifier is generated per app launch and stamped into every scan record.
3	Scan clock	Blank while idle. During a scan it shows the elapsed time <code>HH:MM:SS</code> in green.
4	Window controls	Minimize  , maximize/restore  , and close  .
5	Start / Stop badge	Starts the scan workflow (contact-quality check, then scan) or stops the running scan. Disabled until the system is ready.
6	Notes	Opens the Session Notes editor (§6). Always available.
7	History	Opens Scan History — review, replay, export or delete past scans (§7). Disabled during a scan.
8	Settings	Opens the Settings panel (§8). Disabled during a scan.

#	Element	What it does
9	LEFT / RIGHT panels	Large live readouts of BFI and BVI per side. -- means no valid value (no data yet, or the value is outside the displayable 0–10 range).
10	Plots	One plot per side showing the side-average BFI (white) and BVI (blue) traces.

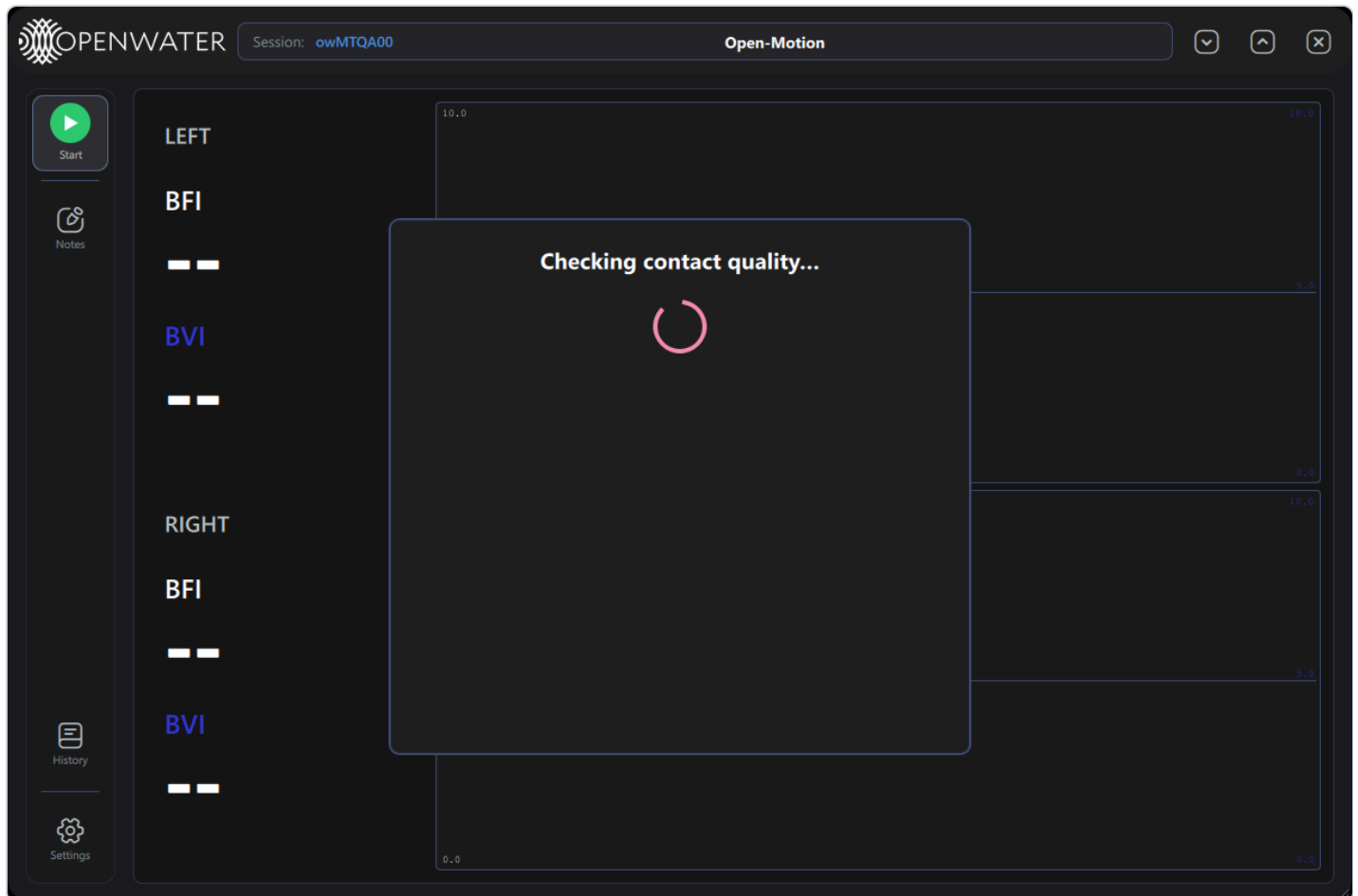
Additional window behaviors:

- **Move** the window by dragging anywhere on the dark header bar.
 - **Resize** using the diagonal grip in the bottom-right corner (minimum size 800 × 600).
 - **Close while busy:** if you press  while a scan or check is running, the app does not exit immediately — a yellow toast warns e.g. "Scan in progress. Click X again to cancel and exit." Pressing  again within 5 seconds cancels the work and exits.
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4. Running a scan

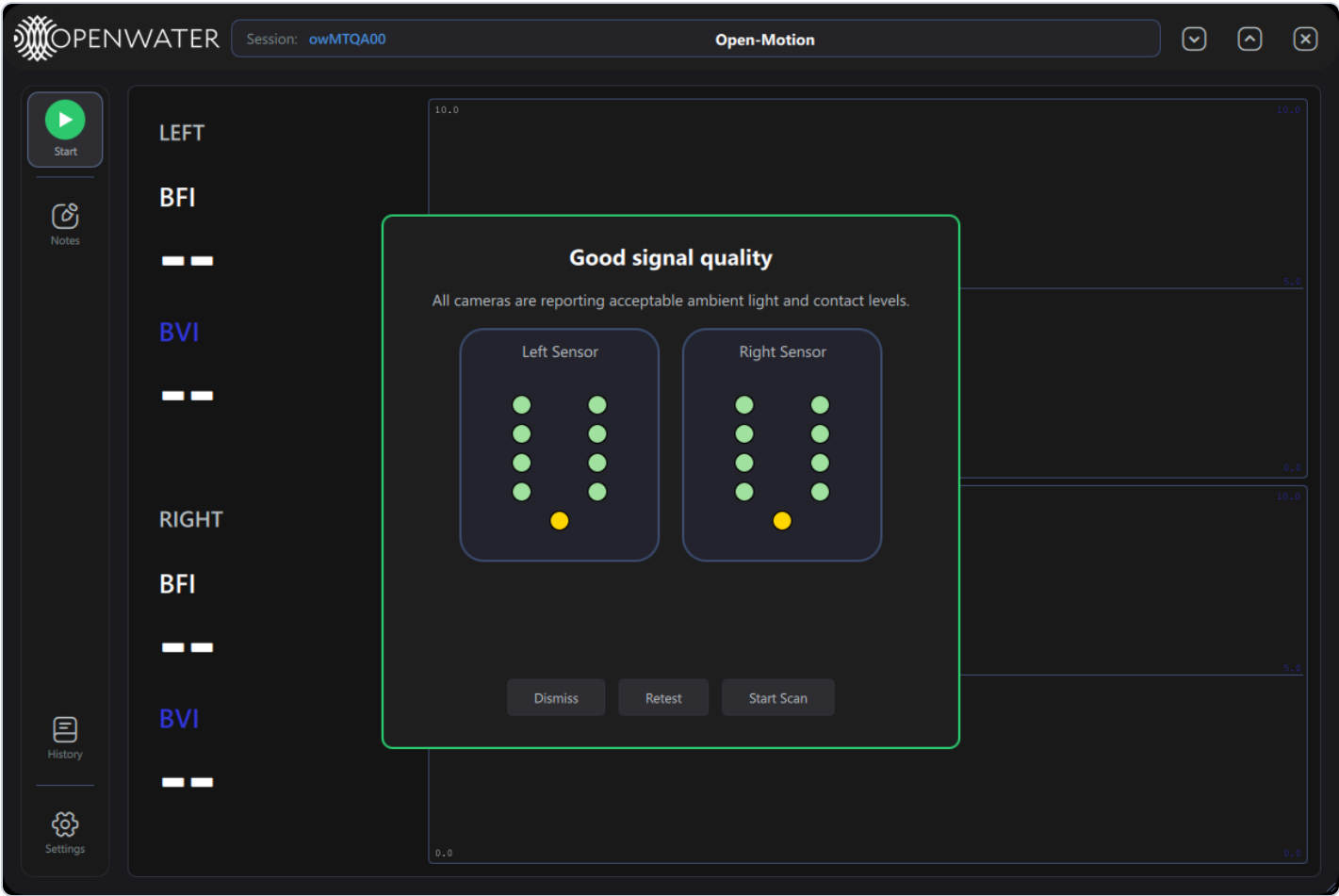
Step 1 — Press Start

Press the green **Start** badge. Every scan begins with an automatic **contact-quality check** of the sensors against the skin:



"Checking contact quality..." — the system briefly configures all cameras and samples ambient light and skin contact. This typically takes well under a minute.

Step 2 — Review the contact-quality result



Each circle is one camera on that sensor (the gold dot marks the laser aperture). The border color of the panel summarizes the outcome:

Color	Meaning
Green border, "Good signal quality"	All cameras report acceptable ambient light and contact.
Orange border, "Contact check failed"	One or more cameras have a problem — hover over an orange dot for the reason (too much ambient light, or poor skin contact). Re-seat the sensor and retest.
Red border, "Contact Quality Notification"	The check itself could not run (e.g. a sensor dropped out).

Camera dot colors: **green** = good · **grey** = not evaluated / still checking · **orange** = problem detected.

Footer buttons:

Button	Action
Dismiss	Close the dialog without scanning.

Button	Action
Retest	Run the contact-quality check again (after adjusting the sensors).
Start Scan	Proceed to the scan. Available as soon as the check has finished — the operator may proceed even past warnings, but for best data quality resolve orange cameras first.

Step 3 — The scan



A scan a few seconds in: the badge is now a red Stop, and the header clock counts up.



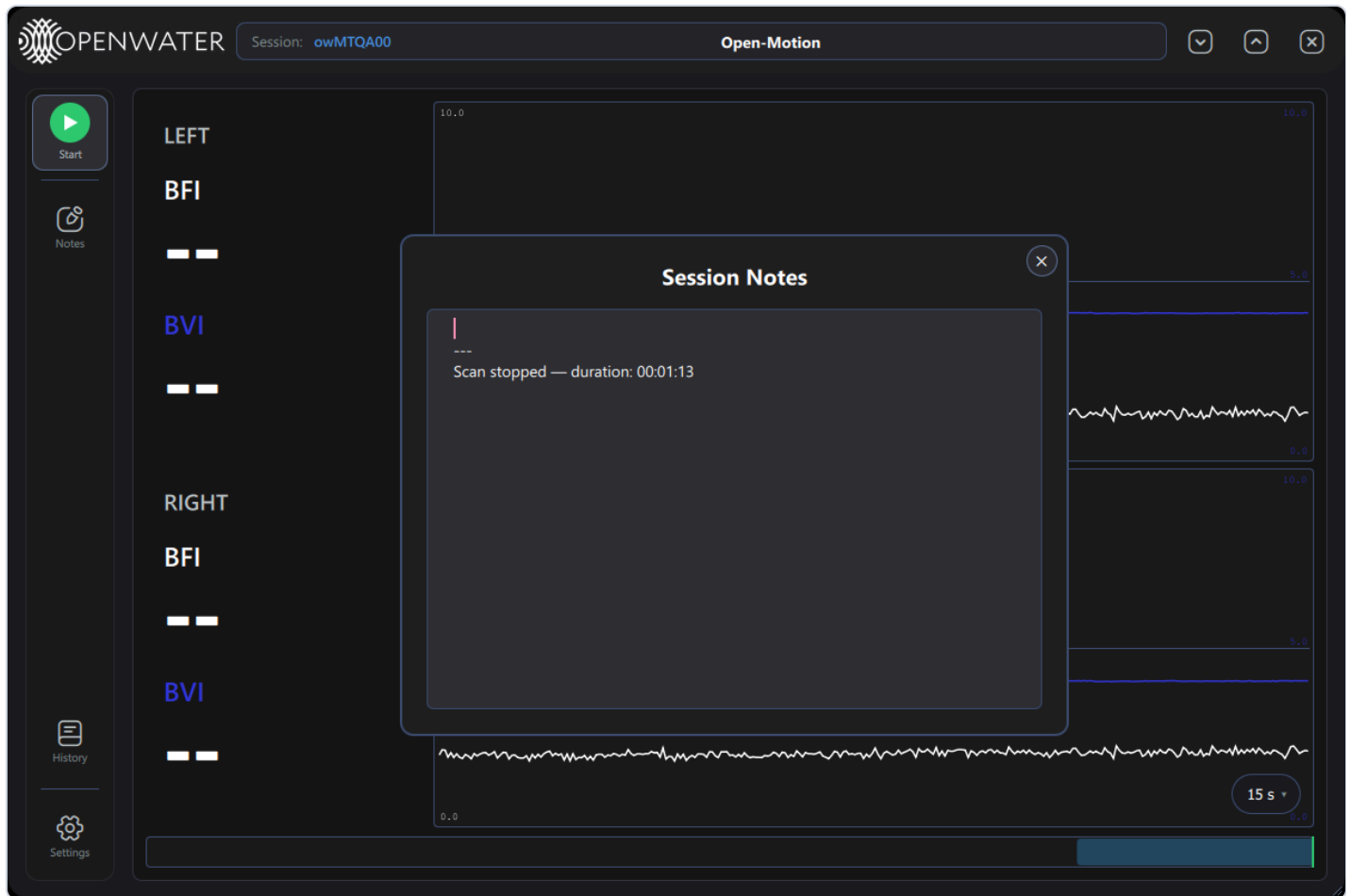
The LEFT/RIGHT panels show live BFI/BVI values; each side's plot draws the BFI (white) and BVI (blue) traces. The floating box in the plot corner is the hover readout — it appears whenever the mouse rests over a plot and shows the values under the cursor.

During the scan:

- The scan is **open-ended**: it runs until you press **Stop** (with a 12-hour ceiling).
- The bottom **timeline bar** (scrubber) and the **15 s** pill control how much history is visible; see §5.
- Press **Space** to open Session Notes with an automatically inserted `[elapsed / wall-clock]` timestamp — type the observation and close (§6).
- If contact quality degrades mid-scan, the contact-quality dialog reappears with a **Stop scan** button (end the scan now) and a **Continue** button. Continue unlocks only after the issue has stayed clear for a couple of seconds.

Step 4 — Stop and annotate

Press the red **Stop** badge. After a few seconds of shutdown the **Session Notes** window opens automatically with a record of the scan:



The stop event and duration are pre-filled (e.g. "Scan stopped — duration: 00:01:13"). Add any observations above the line, then close with  — notes are saved to the scan record ("Note saved." appears in the corner).

All scan data is stored automatically in the application's local database — nothing needs to be exported for the data to be retained.

5. Reading and navigating the plots

The plot area is a DVR: you can rewind during or after a scan without losing the live view.

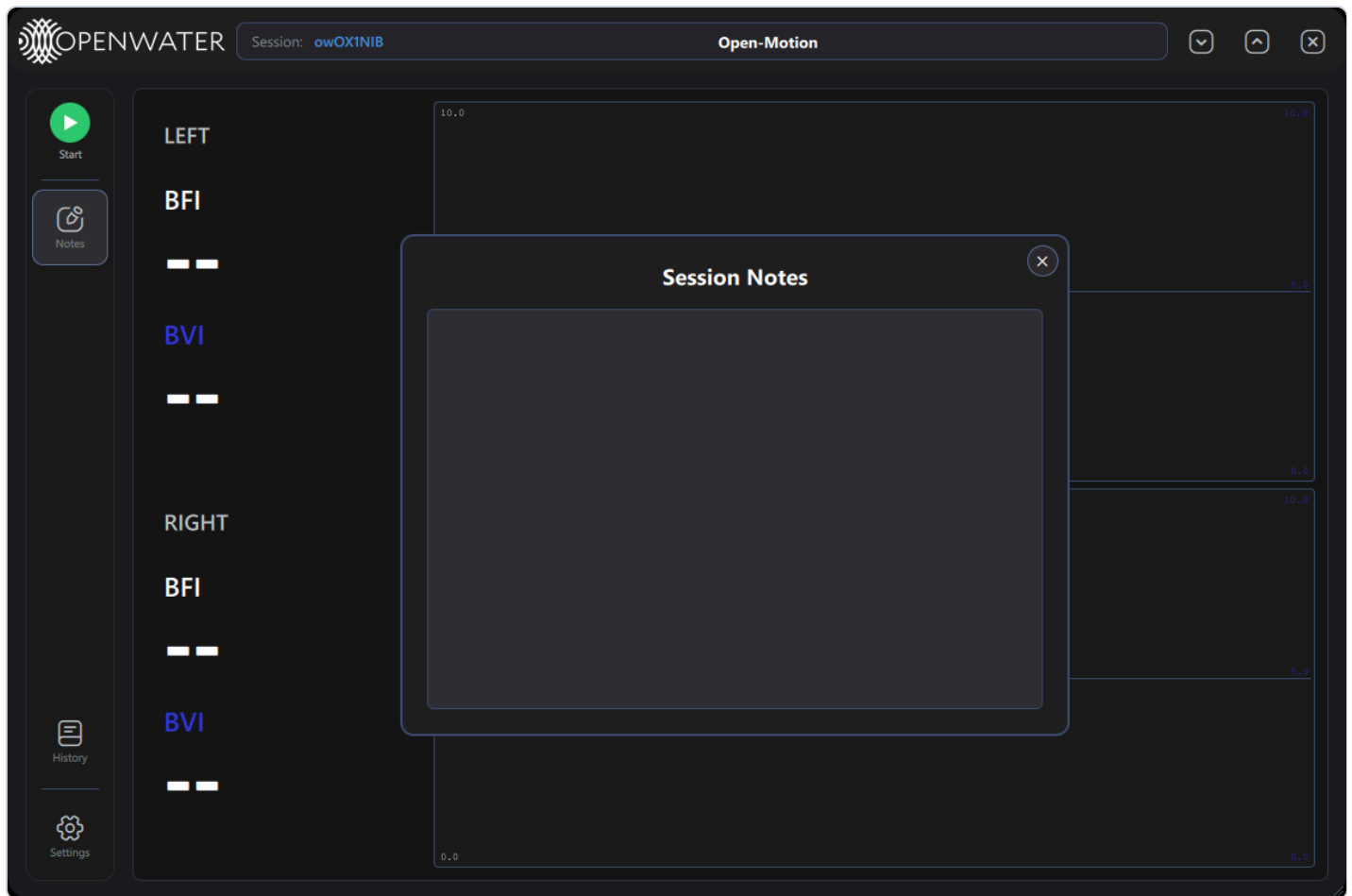
Control	Where	What it does
15 s ▾ pill	bottom-right of the plots	Chooses how many seconds of history are visible: 5 s, 15 s, 30 s, 1 min, 5 min.
Timeline bar	very bottom	Shows the whole scan. Drag the highlighted window to rewind/fast-forward; click anywhere on the bar to jump. Blue window = following live; orange = paused in the past.
Drag on a plot	plot area	Pans back/forward in time (pauses live-follow).
Mouse wheel on a plot	plot area	Zooms the time window in/out around the cursor.
Hover	plot area	Shows the crosshair and the values at the hovered time.
• Back to live pill	top-right of plots	Appears when you have panned into the past during a live scan — click to snap back to the live edge.

Keyboard shortcuts (active when the plot area has focus):

Key	Action
← / →	Pan 1 s back / forward (Shift pans a full window)
+ / ↑	Zoom in
- / ↓	Zoom out
0	Reset the window and return to live
Home / End	Jump to scan start / return to live edge
Space	Open Session Notes with a timestamp (during a scan)
Esc	Close the current dialog

The vertical axis is fixed at the configured bounds (default 0–10 for BFI and BVI; see Settings → Manual Plot Bounds). Values outside the displayable range show as -- in the numeric panels.

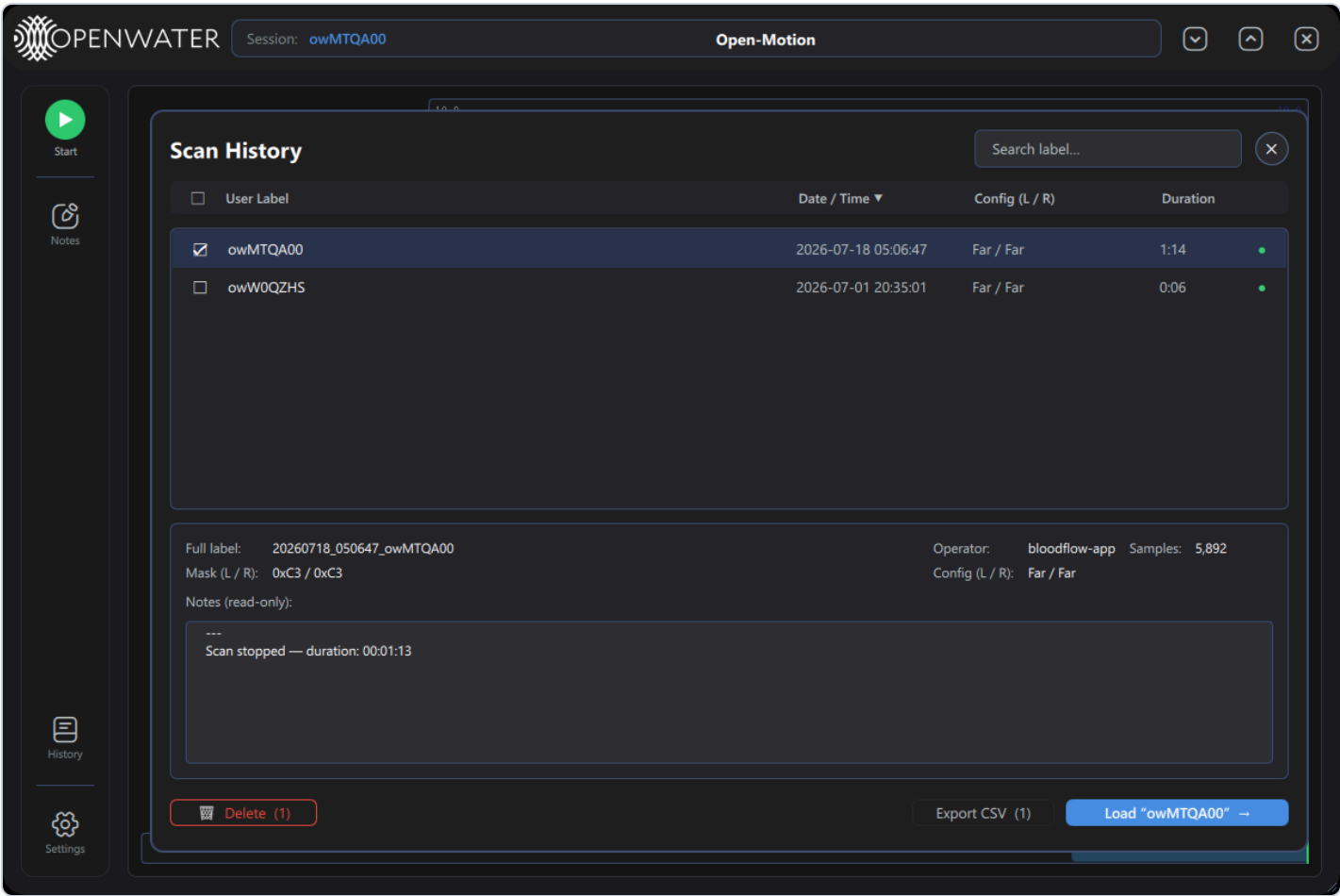
6. Session Notes



- Open at any time with the **Notes** button, or automatically after every scan, or with **Space** during a scan (which inserts a timestamp line for you).
 - Everything you type is saved to the current session's scan record when the window closes (, clicking outside, or `Esc`) — a "Note saved." toast confirms.
 - Notes are visible later in Scan History's detail pane (read-only there).
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7. Scan History

Press **History** to review past scans.



The list shows your recorded scans. Times in the **Date / Time** column are shown in UTC.

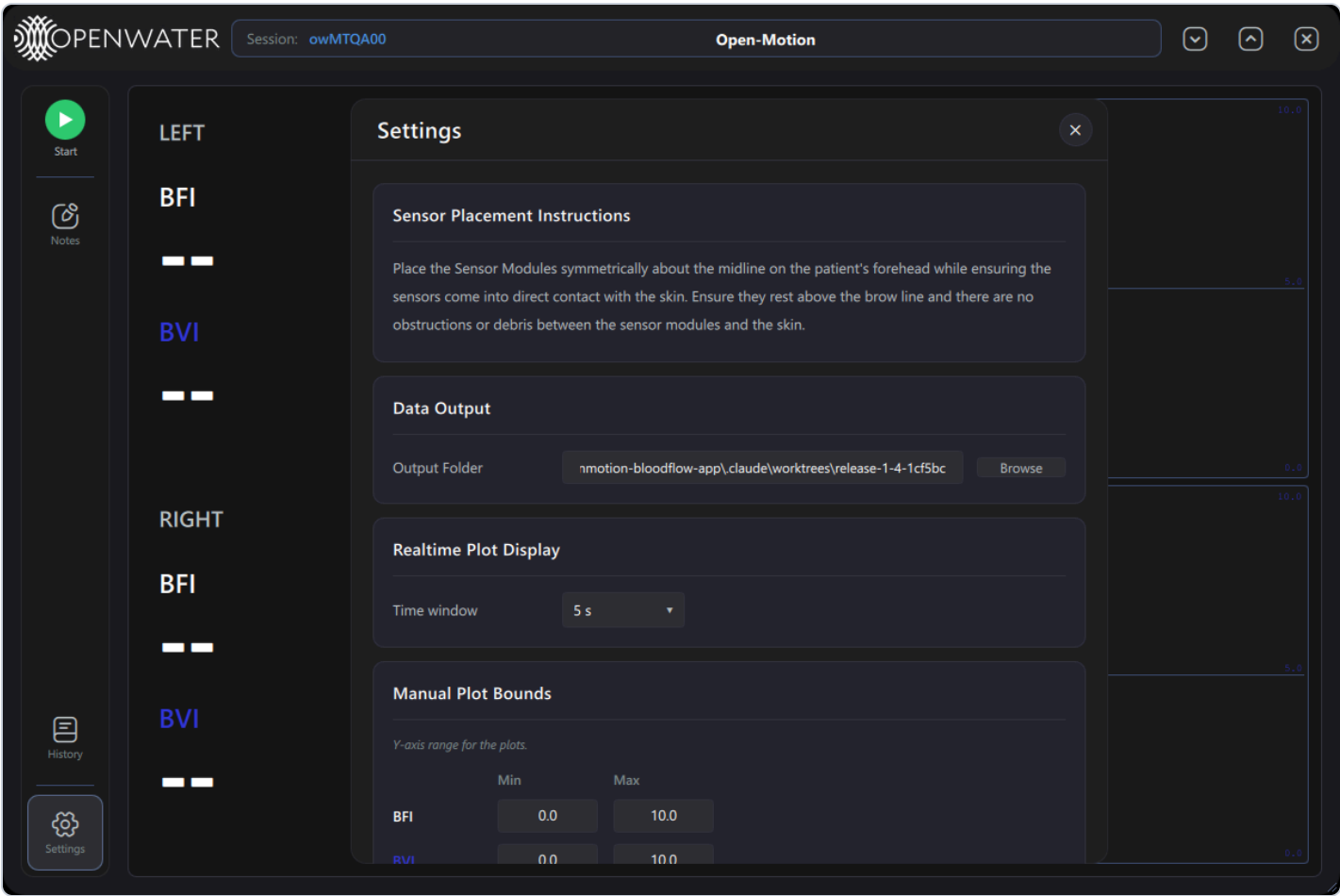
Control	What it does
Search label...	Filters the list by label text.
Column headers	Click <i>User Label</i> , <i>Date / Time</i> , <i>Config (L/R)</i> or <i>Duration</i> to sort; click again to reverse (▲/▼).
Row click	Selects a scan and fills the detail pane below (full label, operator, sample count, camera masks, notes).
Checkbox (per row / header)	Marks scans for Delete / Export. The header checkbox selects all.
Status dot (right edge)	Green ● = complete scan. Amber ⚠ = interrupted scan (cannot be replayed or exported).
🗑 Delete (N)	Permanently deletes the checked scans. Password-protected and irreversible — a "Confirm Delete" prompt asks for a password before anything is removed.

Control	What it does
Export CSV (N)	Exports the checked scan(s) to CSV. One scan opens a save-file dialog; several scans ask for a folder.
Load "label" →	Loads the selected scan into the plot viewer for replay.
×	Closes History.

Replay: after *Load*, the plot area shows the recorded scan with a "Viewing $\langle label \cdot date \rangle$ " badge and a red "←
Back to live scan" pill for returning to the live view. All plot navigation (timeline, zoom, hover) works identically on replays.

8. Settings

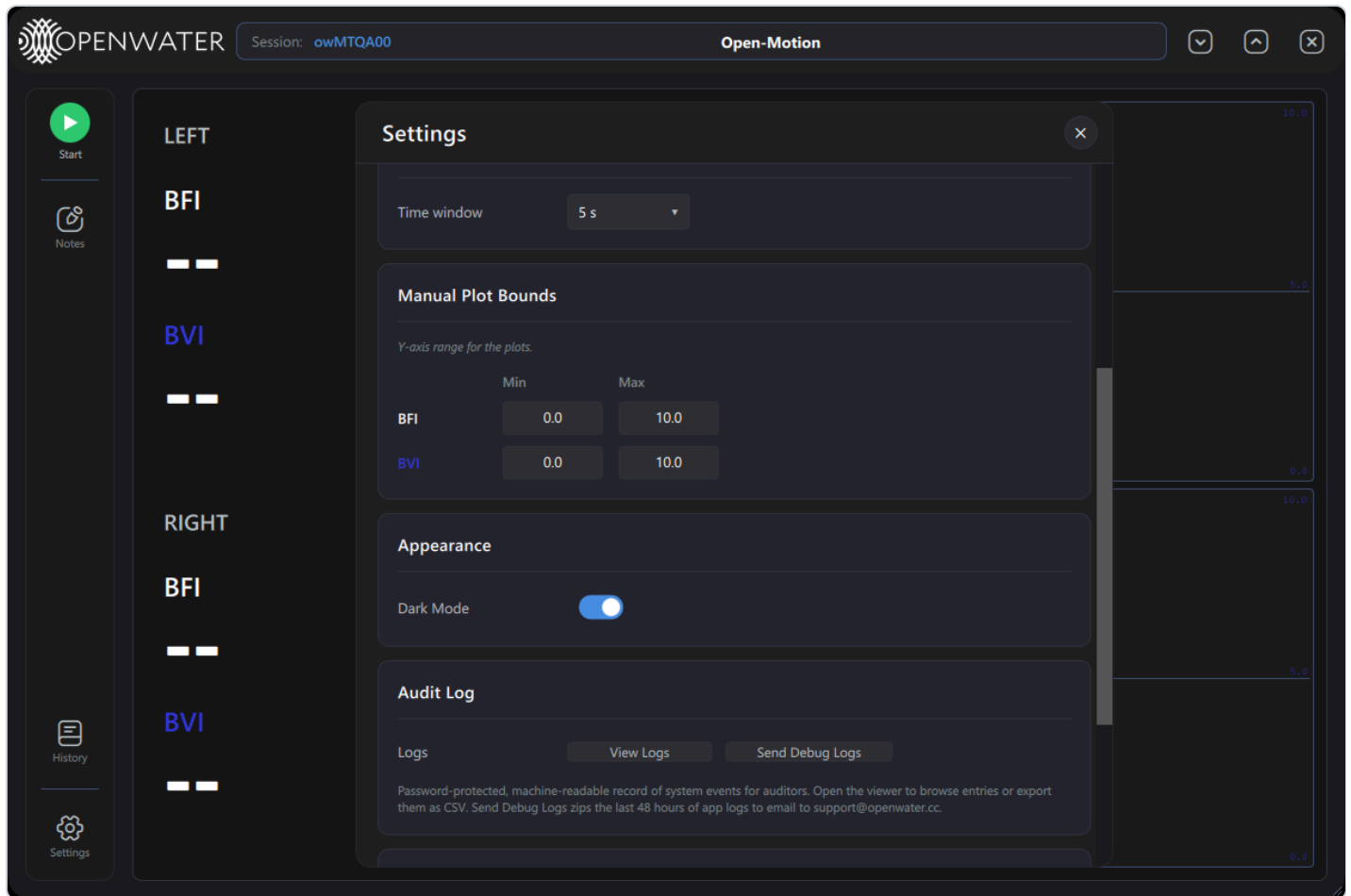
Press **Settings**. The panel contains the following cards (scroll for more):



Sensor Placement Instructions — reference text: place the sensor modules symmetrically about the midline on the patient's forehead, above the brow line, in direct skin contact with no obstructions.

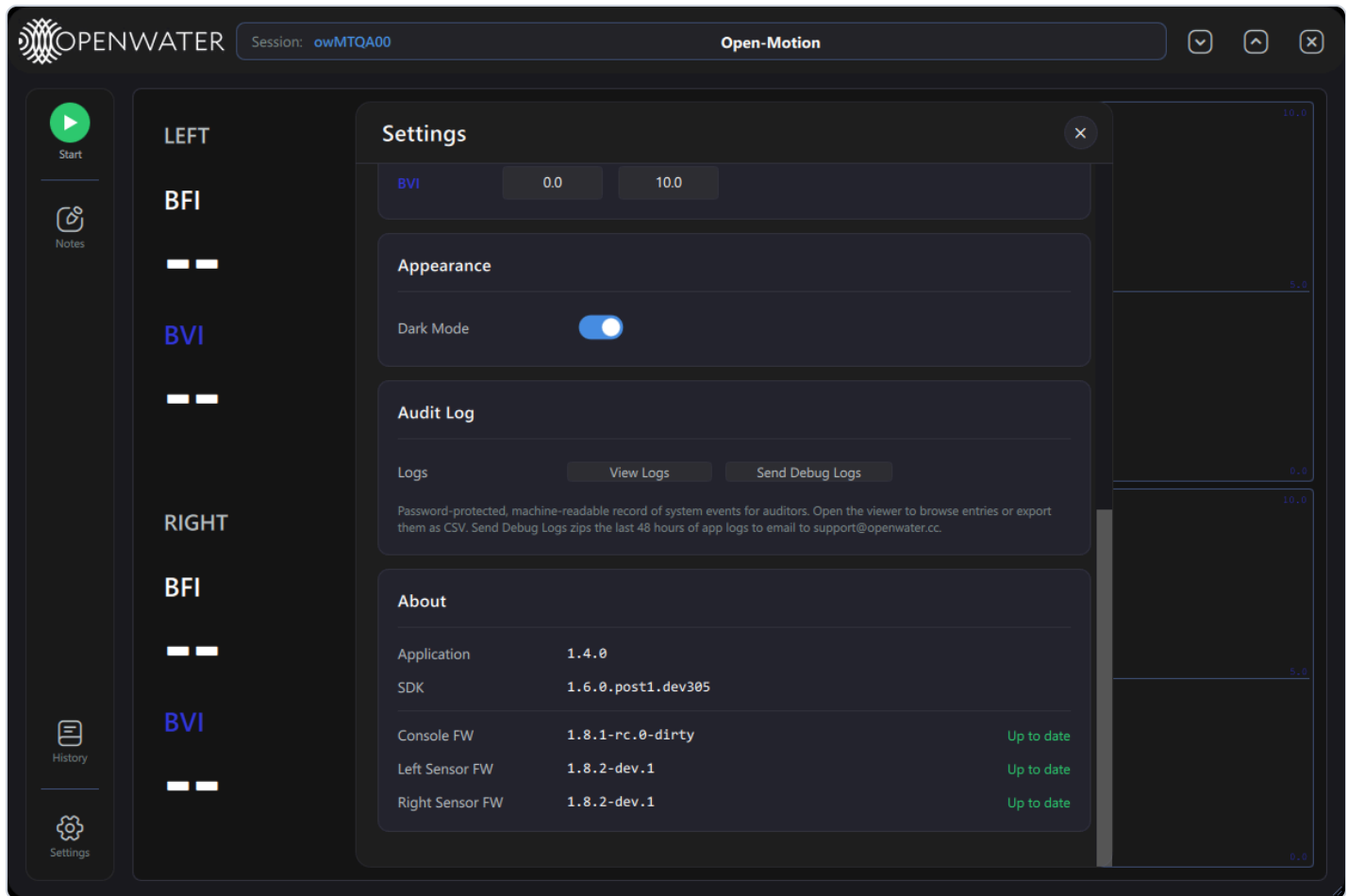
Data Output — where the application stores its data (scan database, logs, exports). *Output Folder* shows the current location; **Browse** selects a different folder.

Realtime Plot Display — *Time window* sets the default plot history length.



Manual Plot Bounds — the fixed Y-axis range used by the plots and numeric panels (Min/Max for BFI and BVI; defaults 0.0–10.0).

Appearance — *Dark Mode* switch. The entire interface re-themes immediately (a light theme is available for bright environments).



Audit Log — a machine-readable record of system events for auditors:

Button	What it does
View Logs	Opens the audit-log viewer. Password-protected (for auditors and Openwater support).
Send Debug Logs	Packages the last 48 hours of application logs into a zip under the data folder, opens its location in Explorer, and shows instructions to email it to support@openwater.cc . Use this when reporting a problem.

About — versions of the application, SDK, console firmware and both sensors' firmware.

Closing Settings ( or **Esc**) saves any changes.

9. Alerts and errors

9.1 Toast notifications

Short status messages appear in the bottom-right corner, color-coded: green = success, yellow = warning, red = error, blue = information. They dismiss themselves after a few seconds (hovering pauses the countdown); some carry an  to dismiss manually. Safety-critical toasts (see below) stay on screen.

Notable messages:

Message	Meaning / action
"Note saved."	Session notes were stored.
"Could not start scan — the previous step is still finishing. Please press Start again."	The system was still busy; wait a few seconds and press Start again.
"Laser safety warning detected. Please restart your console. If this error persists, please contact support." (red, persistent)	The console's hardware laser-safety monitor tripped. Power-cycle the console; contact support if it recurs.
"Laser safety system tripped. Scan will be cancelled in 5 seconds."	A safety trip occurred mid-scan; the scan stops automatically.
Console / sensor "not detected" warnings at startup	A device did not enumerate within the expected time — check cables/power.

9.2 Critical Error dialog

Show-stopping conditions raise a blocking **Critical Error** dialog with an error code, explanation, and a suggested action. Buttons:

Button	What it does
Copy details	Copies the full error report to the clipboard.
Send Bug Report to Openwater	Emails the session log and error context to Openwater support (opens your mail client if direct send is unavailable).
► Details	Expands the technical detail block.
Dismiss	Closes the dialog (shows the next queued error, if any).

9.3 Error code reference

Code	Title	Meaning (summary)	Suggested action
E-101	Sensor self-check failed	A sensor's internal device (mux, IMU, camera, FPGA) failed its power-on self-check.	Power-cycle the sensor and reconnect; if persistent, the sensor needs service.
E-102	Sensor self-check unreadable	The sensor connected but did not report its self-check result.	Power-cycle and reconnect; update firmware or send a bug report if persistent.
E-103	Console initialization failed	The console's laser-power configuration could not be applied.	Power-cycle the console and reconnect; send a bug report if persistent.
E-105	Camera power-on failed	The sensor could not power its cameras during initialization.	Power-cycle the sensor and reconnect; service may be required.
E-201	Laser safety monitor unresponsive	The safety monitor stopped reporting a known-good state; the laser was shut off as a precaution.	Power-cycle the system. Do not scan until this clears.
E-202	Laser safety trip	The safety monitor tripped during a scan; the laser was shut off and the scan stopped.	Remove any obstruction, let the system settle, start a new scan.
E-301	Scan aborted before start	A precondition check failed before the laser fired.	Resolve the reported issue and start again.
E-302	Scan could not start	The system refused a new scan, usually because the previous one is still finishing.	Wait a few seconds and try again.
E-303	Camera data lost during scan	Every camera stopped delivering data; the scan was stopped. Data captured before the loss was saved.	Check sensor cables/power, reconnect, start a new scan.

10. Data storage

All application data lives under the configured output folder (Settings → Data Output):

Location	Contents
logs/	One application log file per launch.
data/scans.db	The scan database: all scan data, sessions and notes.
data/debug-bundles/	Zips created by <i>Send Debug Logs</i> .

Scan data is retained in the database automatically; CSV files are only created when you explicitly export from Scan History.

11. Getting help

1. Use **Settings** → **Send Debug Logs** to package the recent logs.
 2. Email the zip to **support@openwater.cc** with a description of the issue, or use **Send Bug Report to Openwater** directly from a Critical Error dialog.
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